

# A Calc

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## 1 Taylor Series With Remainder

$$\begin{aligned}F(x) &= f(x) - \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i \\F'(x) &= f'(a) - \sum_{i=0}^{n-1} \frac{f^{(i+1)}(a)}{i!} (x-a)^i \\F^{(k)}(x) &= f^{(k)}(x) - \sum_{i=0}^{n-k} \frac{f^{(i+k)}(a)}{i!} (x-a)^i.\end{aligned}$$

Note  $F^{(k)}(a) = 0$  for  $k = 0, 1, \dots, n$ . Suppose  $F^{(n+1)}(x)$  attains a maximum of  $M$  on the interval  $[a, b]$ . We have  $F^{(n+1)}(x) \leq M$  for all  $x \in [a, b]$ , so  $\int_a^x F^{(n+1)}(x) dx \leq M(x-a)$ , or  $F^{(n)}(x) = F^{(n)}(x) - F^{(n)}(a) \leq M(x-a)$ . We may apply this process inductively, so  $F(x) \leq \frac{M}{(n+1)!} (x-a)^{n+1}$  for all  $x \in [a, b]$ .  $F$  is the difference of  $f$  and a polynomial of degree  $n$ , so the function  $F^{(n+1)}$  is the same as  $f^{(n+1)}$ . Hence for any  $x \in [a, b]$ ,

$$f(x) = \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i + R,$$

where  $R \leq \frac{M(x-a)^{n+1}}{(n+1)!}$ .

Book's solution: Define

$$F(x) = f(b) - \sum_{i=0}^n \frac{f^{(i)}(x)}{i!} (b-x)^i - \frac{P}{(n+1)!} (b-x)^{n+1}.$$

$F(a) = F(b) = 0$  and

$$\begin{aligned}
F'(x) &= -f'(x) - \sum_{i=1}^n \frac{f^{(i+1)}(x)(b-x)^i - f^{(i)}(x) \cdot i(b-x)^{i-1}}{i!} + \frac{P}{n!}(b-x)^n \\
&= -f'(x) - \sum_{i=2}^{n+1} \frac{f^{(i)}(x)(b-x)^{i-1}}{(i-1)!} + \sum_{i=1}^n \frac{f^{(i)}(x)(b-x)^{i-1}}{(i-1)!} + \frac{P}{n!}(b-x)^n \\
&= -f'(x) + f'(x) - \frac{f^{(n+1)}(x)(b-x)^n}{n!} + \frac{P}{n!}(b-x)^n \\
&= -\frac{f^{(n+1)}(x)(b-x)^n}{n!} + \frac{P}{n!}(b-x)^n.
\end{aligned}$$

By Rolle's Theorem, there exists some  $c \in [a, b]$  for which  $\frac{P}{n!}(b-c)^n = \frac{f^{(n)}(c)}{n!}(b-c)^n$ , or  $P = f^{(n+1)}(c)$ . It follows that we can bound the error of  $f(b)$  from  $f(a)$  with the maximum of  $f^{(n+1)}(x)$  over  $[a, b]$ .

We essentially have the stronger result:

$$f(x) = \sum_{i=0}^n \frac{f^{(i)}(x)}{i!}(x-a)^i + \frac{f^{(n+1)}(\zeta)}{(n+1)!}(x-a)^{n+1},$$

where  $a < \zeta < x$ .

Error for  $\arctan(x)$ .  $\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots \pm x^{2k} \mp x^{2k+2}P$ , where  $P = \frac{1}{1+x^2} < 1$ , so

$$\begin{aligned}
\arctan(x) &= \int_0^x (1 - x^2 + x^4 - x^6 + \dots + x^{2k} - x^{2k+2}P)dx \\
&= x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \dots \pm \frac{1}{2k+1}x^{2k+1} \pm \int_0^x x^{2k+2}Pdx.
\end{aligned}$$

Now,

$$\begin{aligned}
\int_0^x x^{2k+2}Pdx &< \int_0^x x^{2k+2}dx = \frac{1}{2k+3}x^{2k+3}. \\
\arctan(x) &= x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \dots \pm \frac{1}{2k+1}x^{2k+1} + R,
\end{aligned}$$

where  $|R| < \frac{1}{2k+3}x^{2k+3}$ .

Other applications of Rolle's Theorem. Consider the function

$$G(x) = f(b) - \sum_{i=0}^n \frac{f^{(i)}(x)}{i!}(b-x)^i - \frac{b-x}{b-a}R.$$

Note that  $f(a) = f(b) = 0$ .

$$\begin{aligned}
G'(x) &= -f'(x) - \sum_{i=1}^n \frac{f^{(i+1)}(x)(b-x)^i - f^{(i)}(x)i(b-x)^{i-1}}{i!} + \frac{R}{b-a} \\
&= -f'(x) - \sum_{i=1}^n \frac{f^{(i+1)}(x)(b-x)^i}{i!} + \sum_{i=1}^n \frac{f^{(i)}(x)(b-x)^{i-1}}{(i-1)!} + \frac{R}{b-a} \\
&= -f'(x) - \sum_{i=1}^n \frac{f^{(i+1)}(x)(b-x)^i}{i!} + \sum_{i=0}^{n-1} \frac{f^{(i+1)}(x)(b-x)^i}{i!} + \frac{R}{b-a} \\
&= -\frac{f^{(n+1)}(x)(b-x)^n}{n!} + \frac{R}{b-a},
\end{aligned}$$

so by Rolle's there exists  $\zeta \in (a, b)$  for which  $R = \frac{f^{(n+1)}(\zeta)(b-\zeta)^n(b-a)}{n!}$ . Write  $\zeta = a + (b-a)\theta$  for  $0 < \theta < 1$ , so  $R = \frac{f^{(n+1)}(\zeta)(b-a)^{n+1}(1-\theta)^n}{n!}$ .

Alternative approach to result with integrals.  $f(x) - f(a) = \int_0^{x-a} f'(x-t)dt$ . By Integration by Parts,

$$f(x) - f(a) = (x-a)f'(a) + \int_0^{x-a} tf''(x-t)dx.$$

In general, we may write:

$$f(x) - f(a) = \sum_{i=1}^n \frac{f^{(i)}(a)}{i!} (x-a)^i + R,$$

where

$$R = \int_0^{x-a} \frac{f^{(n+1)}(x-t)}{n!} t^n dt.$$

## 2 Indeterminate Forms

Form 0/0. Consider the function

$$G(x) = \frac{f(b) - f(a)}{\phi(b) - \phi(a)} (\phi(x) - \phi(a)) - (f(x) - f(a)).$$

$$G(a) = G(b) = 0.$$

$$G'(x) = \frac{f(b) - f(a)}{\phi(b) - \phi(a)} \phi'(x) - f'(x)$$

By Rolle's Theorem, there exists  $a < \zeta < b$  for which

$$\frac{f'(\zeta)}{\phi'(\zeta)} = \frac{f(b) - f(a)}{\phi(b) - \phi(a)}.$$

Now let  $b = x$  and  $f(a) = \phi(a) = 0$ . Then

$$\lim_{x \rightarrow a} \frac{f(x)}{\phi(x)} = \lim_{\zeta \rightarrow a} \frac{f'(\zeta)}{\phi'(\zeta)}.$$

Form  $\infty/\infty$ .

$$\lim_{x \rightarrow 0} \frac{f(x)}{\phi(x)} = \lim_{x \rightarrow 0} \frac{1/\phi(x)}{1/f(x)} = \lim_{x \rightarrow 0} \frac{-\phi'(x)/\phi(x)^2}{-f'(x)/f(x)^2} = \left( \lim_{x \rightarrow 0} \frac{f(x)}{\phi(x)} \right)^2 \lim_{x \rightarrow 0} \frac{\phi'(x)}{f'(x)}.$$

Hence, if  $L = \lim_{x \rightarrow \infty} f(x)/\phi(x)$ ,  $M = \lim_{x \rightarrow \infty} \phi'(x)/f'(x)$ , then  $L = L^2 M$ . Either  $L = 0$  or  $LM = 1$ ...? If  $L, M$  are nonzero,

$$\lim_{x \rightarrow \infty} \frac{f(x)}{\phi(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{\phi'(x)}$$

Textbook attempt. Take  $a = \infty$ . Consider

$$\frac{f(x)}{\phi(x)} = \frac{f'(\zeta)}{\phi'(\zeta)} \cdot \frac{1 - \phi(a)/\phi(x)}{1 - f(a)/f(x)}$$

for  $a < \zeta < x$ . Take  $a$  sufficiently large so that  $\frac{f'(a)}{g'(a)}$  is at most  $\epsilon_1$  away from its limit  $A$ . Then  $\frac{f'(\zeta)}{\phi'(\zeta)}$  will also be at most  $\epsilon_1$  away from  $A$ . Take  $x$  sufficiently large so that  $\frac{1 - \phi(a)/\phi(x)}{1 - f(a)/f(x)}$  is at most  $\epsilon_2$  away from 1. We have  $\frac{f(x)}{g(x)} = (1 + \eta_2)(A + \eta_1)$ , where  $|\eta_1| < \epsilon_1$  and  $|\eta_2| < \epsilon_2$ .  $\epsilon_1, \epsilon_2$  can be taken to be arbitrarily small, so

$$\lim_{x \rightarrow \infty} \frac{f(x)}{\phi(x)} = A = \lim_{x \rightarrow \infty} \frac{f'(x)}{\phi'(x)}.$$

Now suppose  $a$  is finite. Note

$$\lim_{y \rightarrow \infty} \frac{f(a + 1/y)}{\phi(a + 1/y)} = \lim_{y \rightarrow \infty} \frac{-\frac{1}{y^2} f'(a + 1/y)}{-\frac{1}{y^2} \phi'(a + 1/y)} = \lim_{x \rightarrow a} \frac{f'(x)}{\phi'(x)}.$$

Infinitesimals. Define an infinitesimal  $\alpha$  to be a function of some variable  $t$  such that  $\alpha \rightarrow 0$  as  $t \rightarrow 0$ . If  $\alpha_1, \alpha_2, \dots, \alpha_n$  are infinitesimals, then  $k_1\alpha_1 + k_2\alpha_2 + \dots + k_n\alpha_n$  is also an infinitesimal.  $\alpha_1\alpha_2 \dots \alpha_n$  is an infinitesimal. Say  $\alpha, \beta$  are infinitesimals of the same order of  $\lim_{t \rightarrow 0} \alpha/\beta$  is a nonzero constant. Write  $\alpha \sim \beta$ . Then  $\sim$  is an equivalence relation.

Say that an infinitesimal  $\alpha$  is of order  $n > 0$  if  $\alpha$  and  $t^n$  are of the same order. If this is the case,  $\lim_{t \rightarrow 0} \alpha/t^n = k$ , so

$$\lim_{t \rightarrow 0} \frac{\alpha - kt^n}{t^n} = 0,$$

or that  $\alpha - kt^n = t^n \epsilon$ , or  $\alpha = kt^n + t^n \epsilon$ . We may rewrite this as  $\alpha = t^n(k + \epsilon)$ , where  $\epsilon$  is an infinitesimal. Alternatively, write  $\alpha = \alpha_1(1 + \epsilon)$ .

Consider the ratio of two infinitesimals  $\alpha, \beta$ .

$$\lim_{t \rightarrow 0} \frac{\alpha}{\beta} = \lim_{t \rightarrow 0} \frac{\alpha_1(1 + \epsilon_1)}{\beta_1(1 + \epsilon_2)} = \lim_{t \rightarrow 0} \frac{\alpha_1}{\beta_1}.$$

Sum of infinitesimals. Let  $\alpha_1, \alpha_2, \alpha_3, \dots$  be a sequence of positive infinitesimals such that  $\alpha_1 + \alpha_2 + \alpha_3 + \dots$  converges. Then the sum of the infinitesimals at  $t = 0$  is unaltered by replacing each  $\alpha_i$  by its principal part  $\beta_i$ .

Consider  $\alpha_1, \alpha_2, \dots, \alpha_n$  and let  $\alpha_i = \beta_i + \beta_i \epsilon_i$ . Let  $\gamma$  denote the largest absolute value of the  $\epsilon_i$ . Then  $-\gamma \leq \epsilon_i \leq \gamma$ , or  $-\gamma \beta_i \leq \beta_i \epsilon_i \leq \gamma \beta_i$ . Summing gives  $-\gamma \sum \beta_i \leq \sum \beta_i \epsilon_i \leq \gamma \sum \beta_i$ .  $\gamma \rightarrow 0$  and  $\sum \beta_i$  is bounded, so  $\sum \beta_i \epsilon_i \rightarrow 0$ . Hence,  $\sum \alpha_i = \sum \beta_i$  as  $n \rightarrow \infty$ .